# Devika Mukherjee

XR/VR | Wearable Haptics | Human-AI Interaction

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## PROFILE

HCI researcher building and testing mixed and virtual reality systems, with a focus on wearable haptics and human-AI interaction. My first-author CHI 2026 paper introduced a haptic belt that warns MR users of lower-body collisions, tested with 35+ participants. I prototype in Unity (C#), run user studies end to end, and analyse behavioural and time-series data in Python and R. I want to research adaptive XR assistance that helps people live independently while keeping them in control.

## SKILLS

**XR & Interaction Design:** MR/VR prototyping in Unity (C#), wearable haptics, cross-modal feedback (haptic, spatial, visual), LLM-driven and agentic XR, accessible design
**Data & Analysis:** behavioural and time-series data, real-time alert thresholds, combining task metrics with self-report measures
**User Research:** controlled experiments, mixed methods, co-creation workshops, interviews, usability and workload measures (SUS), journey mapping, systematic reviews
**Human-AI Interaction:** trust and reliance, user control over AI changes, auditing LLM output (accuracy, hallucination, stability)
**Programming:** C#, Python (pandas, NumPy, SciPy), R, Git, Jupyter, Plotly, Streamlit, ggplot2, LLM APIs
**Languages:** English (C1+, IELTS 8.0)

## EDUCATION

09/2024 – 09/2025 **MSc Human-Computer Interaction, University of Birmingham**
09/2020 – 09/2024 **Bachelor of Technology in Biotechnology, Amity University**

## PUBLICATIONS

**Mukherjee, D., Abdlkarim**, D., Giunchi, D., Di Luca, M., Ofek, E. (2026).
**"Belt and Whistles: Adding Lower Body Collision Awareness for MR Experiences."**
ACM CHI Conference on Human Factors in Computing Systems (CHI 2026), Barcelona, Spain.
https://dl.acm.org/doi/10.1145/3772318.3793143

**Mukherjee, D.**
**"Karyotyping and Down Syndrome: Insights from Cytogenetic Analysis and Patient Data."**
International Journal for Research in Applied Science & Engineering Technology (IJRASET), ISSN: 2321-9653; SJ Impact Factor: 7.538; Volume 12, Issue V, May 2024. https://doi.org/10.22214/ijraset.2024.62206

***Submitted / In Review***
**"Continuous World Generation: Real-Time, LLM-Driven Voxel Biome Synthesis for Immersive Virtual Environments."**
Submitted to IEEE, 2027

***Submitted / In Review***
**"Worlds That Think With Us: Toward Symbiotic AgenticXR."**
Submitted to IEEE, 2027

## WORK EXPERIENCE

**Research Associate**, University of Birmingham
Jun 2025 – Present

The Belt Project, VR Lab

- Designed and built a wearable haptic belt prototype and evaluated it in controlled VR studies with 35+ participants, owning the full research cycle from study design and recruitment to analysis and reporting.
- Researching cross-modal, non-verbal feedback (haptic and spatial cues) as a hands-free communication channel in immersive systems, including evaluation of AI-driven interactive systems for reliability and user trust.
- Analysed behavioural, task-performance and user-experience data in Python, combining interaction metrics with self-report measures to produce evidence-based design recommendations.
- Evaluated how wearable haptic feedback communicates spatial and safety information *non-verbally*, examining immersion, perception and task performance for a novel, non-auditory interaction channel, with applications in accessible, context-aware alerting.
- Collaborated with internationally recognised HCI researchers; work resulted in a first-author publication at CHI 2026 (DOI).

Human-Computer Interaction Team Member, DreamcodeVR

- Continuous World Generation: HCI contributor on a system that generates immersive VR content from free-form natural-language requests in real time, including Minecraft-style biomes. Ran a 16-participant user study evaluating intent fidelity, perceived control and usability (SUS = 73.4), assessing how well the system adapted the world to what users actually asked for.
- Worlds That Think With Us: Toward Symbiotic AgenticXR: Leading experiment design and ongoing development for an agentic XR system in which AI proposals must pass a Verification Space (preview, confirmation, repair, rollback) before they change the live scene, so users stay in control of AI-driven adaptation.

Research Collaborator, HCI Fabrication Techniques, Robotics Lab. Computer Science 2026 – Present

- Co-authored a submission to CHI 2027 systematically characterising Laser-Induced Forward Transfer (LIFT) printing of graphene ink across substrates relevant to HCI (tattoo paper, PDMS, PET, TPU-coated nylon, bamboo), including electrical and mechanical characterisation of the printed material.
- Contributed to demonstrator interactive devices built with LIFT-printed graphene, including touch/pressure sensing and an inflatable joystick prototype for motor-disability accessibility.

Research Associate & Associate Project Manager, ATOUCH

Translated research objectives into funding proposals aligned with industry and business priorities, building experience in framing research value for non-research stakeholders.

- Coordinated industry partnerships and stakeholder engagement across academia and business, supporting research aimed at real-world applications.

Freelance Software Development Project, Human-Robot Interaction, Extreme Robotics Lab, UoB

- Built a Unity simulation of robotic battery-disassembly workflows (KUKA KR-16) supporting ML integration, and delivered demo-ready assets for stakeholder presentations.

## RELEVANT PROJECTS

**"Signal or Noise": a clinician triage interface for remote heart failure monitoring under alert overload —** Link

- Built and deployed a decision-support prototype (Python, Streamlit, Plotly, pandas) that reframes alert-threshold selection as a clinician's 'attention budget', translating a user constraint directly into a system parameter.
- Audited an LLM-generated "why-flagged" rationale against structured ground truth (120 generations): **15.8% contained values absent from the input; only 74% were numerically stable** across identical runs, so the deterministic explainer ships as the default — prioritising a trustworthy, reproducible explanation over a more fluent but unreliable AI-generated one.

*Skills: time-series alerting · threshold selection · Streamlit · Plotly · clinical decision support · simulation*

**Reliability Audit of an LLM-Based Voting Advice Application —** GitHub

- Evaluation harness testing whether an LLM can reliably perform party-matching for Voting Advice Applications, using official Wahl-O-Mat data (2025 German federal election) with computed ground truth.
- Factorial design across 25 voter profiles and 6 prompt conditions, measuring accuracy, run-to-run stability, positional bias, and sycophancy — a general-purpose method for testing whether an AI system's output can be trusted to represent what it claims to.
- Python pipeline against a production LLM API: structured outputs, caching, append-only logging, metrics pre-specified before data collection.

*Skills: LLM APIs · structured outputs · factorial design · reliability and bias measurement*

**Earlier Project Work**

- **Better Diet: AI Nutrition Tracking App** — Interviewed and surveyed 50+ participants, built personas and journey maps, and co-designed interface concepts for an AI-driven behaviour-change product through direct participatory research with end users. Case study
- **UX Analytics & Data Visualisation** — Analysed relationships between behavioural performance data and UX metrics in R using Spearman correlation and Wilcoxon tests, communicated as actionable design recommendations. Repository

## REFERENCES


**1. Prof. Daniele Giunchi** — School of Computer Science, University of Birmingham, d.giunchi@bham.ac.uk

**2. Prof. Eyal Ofek** — School of Computer Science, University of Birmingham, e.ofek@bham.ac.uk

**3. Prof. Massimiliano Di Luca** — School of Computer Science, University of Birmingham, m.diluca@bham.ac.uk